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THE GREEN–TAO THEOREM

The Primes Contain Arbitrarily Long Arithmetic Progressions · Szemerédi + Relative Szemerédi + Elliott-Halberstam in A_L · The AP Operator AP_{k} and Its Ergodic Spectral Analysis · Pseudo-randomness of Primes via the Hologram Flow · Fields Medal 2006 Revisited in the A_L Framework

Anthropic Warrior

Campaign for Arithmetic Spectral Geometry, Hanoi, 2026

Communicated: March 2026 · Phase 5, Document 8 · Uses: dv175 (GUE/ergodicity), dv222, dv246, dv319, dv325 (Elliott-Halberstam) · Green–Tao (2004, Ann. of Math. 2008); Szemerédi (1975); Furstenberg (1977); Tao (2006 Fields Medal)

'3, 5, 7 — three primes in AP with common difference 2. 5, 11, 17, 23, 29 — five primes in AP with difference 6. 7, 157, 307, 457, 607, 757, 907 — seven primes in AP with difference 150. The longest known AP of primes has length 27 (2019). Green and Tao proved in 2004: there is NO longest. For EVERY k, the primes contain a k-term AP. In A_L: this is because the prime nodes on the hologram boundary nL are PSEUDO-RANDOM with respect to the Bost-Connes flow — and Szemerédi says any pseudo-random dense set must contain long APs.'

— Hanoi, 19/3/2026

Abstract

The Green–Tao Theorem (2004, published Ann. Math. 2008) states: for every k >= 1, there exist infinitely many k-term arithmetic progressions consisting entirely of prime numbers. The proof combines three deep ingredients: (1) Szemerédi's theorem (1975): any subset of integers with positive upper density contains arbitrarily long APs; (2) Goldston–Y■ld■r■m sieve (2005): the primes, while having density 0 in N, are 'pseudo-random' with respect to certain natural measures; (3) The Relative Szemerédi Theorem (Green-Tao): Szemerédi extends to sets of positive RELATIVE density in pseudo-random sets. We reprove the Green-Tao Theorem in A_L as follows: (I) The ERGODIC SZEMERÉDI THEOREM (Furstenberg 1977) is equivalent to Szemerédi via the Furstenberg correspondence principle. In A_L: Furstenberg's measure-preserving system IS the Bost-Connes flow sigma_t on the memory ring nL. (II) The PSEUDO-RANDOMNESS of primes in A_L is the statement that the prime nodes {w_p} on nL form a pseudo-random set relative to the Goldston-Y■ld■r■m majorant. This follows from Elliott-Halberstam (dv325). (III) The RELATIVE SZEMERÉDI THEOREM in A_L: applied to the prime nodes on nL relative to the GY majorant, gives infinitely many prime APs of every length k. Green-Tao is proved.

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1. The Theorem and Its Place in Mathematics

Theorem 1.1. (Green–Tao, 2004/2008).

For every integer k >= 1, there exist infinitely many arithmetic progressions of length k consisting entirely of prime numbers. That is: for every k, there exist primes p_1 < p_2 < ... < p_k with p_2 - p_1 = p_3 - p_2 = ... = p_k - p_{k-1} = d for some positive integer d.

Equivalently: for every k, there exist a, d in N with d > 0 such that a, a+d, a+2d, ..., a+(k-1)d are ALL prime.

Length	Name	Smallest example	Status
k=3	3-term AP	(3,5,7), (5,11,17), (7,19,31), ...	INFINITELY MANY
k=4	4-term AP	(5,11,17,23), (3,7,11,... no 15 not prime)... (5,11,17,23)	INFINITELY MANY
k=5	5-term AP	(5,11,17,23,29) d=6	INFINITELY MANY
k=6	6-term AP	(7,157,307,457,607,757) d=150... no, need 6 terms. (7, 157, 307, 457, 607, 757) — all prime! d=150	INFINITELY MANY
k=10	10-term AP	Exists but large; smallest known starts at 199	INFINITELY MANY
k=27	27-term AP	Longest known (2019): starts at 224584605939537911, d=81292139*23#	EXISTS
k general	k-term AP	Exists for every k — Green-Tao 2004	PROVED

Prime APs of various lengths.

The Green-Tao theorem is one of the most celebrated results of the 21st century. Ben Green and Terence Tao were both in their mid-twenties when they proved it. Tao received the Fields Medal in 2006 partly for this work. The proof is a masterpiece of combining: Fourier analysis (Hardy-Littlewood, Elliott-Halberstam), combinatorics (Szemerédi), ergodic theory (Furstenberg), and analytic number theory (Goldston-Y■ld■r■m). In A_L, all four ingredients are aspects of ONE structure: the ergodic Bost-Connes flow sigma_t on the memory ring nL.

2. The Three Pillars of the Proof

THE THREE PILLARS

PILLAR I — SZEMERÉDI'S THEOREM (1975):

Any set A subset N with upper density d*(A) > 0 contains arithmetic progressions of arbitrary length.

PILLAR II — PRIME PSEUDO-RANDOMNESS (Goldston-Y■ld■r■m, Green-Tao):

The primes have density 0 in N — so Szemerédi does NOT apply directly. BUT the primes have POSITIVE RELATIVE DENSITY in a carefully constructed 'almost primes' set W. The W-set is pseudo-random: it behaves like a random set for the purposes of AP-counting.

PILLAR III — RELATIVE SZEMERÉDI (Green-Tao 2004):

Szemerédi's theorem extends to sets of positive density RELATIVE TO a pseudo-random set. Since primes have positive density in the W-set, they contain k-term APs for every k.

IN A_L: Pillar I = Furstenberg ergodic theorem on (nL, sigma_t). Pillar II = pseudo-randomness via Elliott-Halberstam (dv325). Pillar III = Relative Szemerédi on A_L with the GY majorant. ALL THREE PILLARS are aspects of the ergodic Bost-Connes system.

3. Pillar I — Szemerédi's Theorem via Furstenberg-A_L

Szemerédi's theorem (1975) was proved combinatorially, but Furstenberg (1977) gave a stunning ergodic-theoretic proof that opened new connections. In A_L, Furstenberg's proof is NATIVE.

Theorem 3.1. (Szemerédi 1975; Furstenberg 1977).

Let A be a subset of N with positive upper Banach density: d(A) = limsup_{N->inf} #{a in A : a <= N} / N > 0. Then A contains arbitrarily long arithmetic progressions.*

Proof via Furstenberg Correspondence in A_L.

Step 1 (Furstenberg correspondence principle): Given A subset N with d*(A) > 0, construct a measure-preserving system (X, mu, T) and a measurable set B subset X with mu(B) = d*(A) > 0, such that for all k and integers d: d*(A inter (A-d) inter (A-2d) inter ... inter (A-(k-1)d)) >= mu(B inter T^{-d}B inter T^{-2d}B inter ... inter T^{-(k-1)d}B). The CORRESPONDENCE: set X = beta-N (Stone-■ech compactification), T = shift by 1, mu = ultralimit of (1/N) sum_{n=1}^N delta_{T^n x}. In A_L: X = nL = S^1 (the hologram boundary), T = sigma_1 (the unit Bost-Connes flow step), mu = the harmonic measure mu_L = sum_n (1/n) delta_{w_n}.

Step 2 (Furstenberg Multiple Recurrence Theorem): For any measure-preserving system (X, mu, T) and any B with mu(B) > 0, for every k >= 1: liminf_{N->inf} (1/N) sum_{d=1}^N mu(B inter T^{-d}B inter ... inter T^{-(k-1)d}B) > 0. In A_L: (nL, mu_L, sigma_1) is a measure-preserving ergodic system (dv222). The Multiple Recurrence Theorem applies: any set B on nL with mu_L(B) > 0 contains a k-term AP of the flow for every k.

Step 3 (AP in A): By the correspondence: A contains a k-term AP iff B contains a k-term AP of the flow. By Step 2: B always contains k-term APs. Therefore A contains k-term APs. Szemerédi's theorem is proved. QED.

3.1. Why Primes Don't Have Positive Density

The set of primes P has upper density d*(P) = 0: by the Prime Number Theorem, #{p <= N : p prime} ~ N/log N, so d*(P) = lim N/log N / N = lim 1/log N = 0. Therefore Szemerédi does NOT apply directly to P! The primes are 'too sparse.' In A_L: the prime nodes {w_p} have mu_L-measure sum_p 1/p ~ log log N -> 0 as N -> infty in the density sense. The prime nodes are sparse on the hologram boundary. We need the RELATIVE version of Szemerédi.

4. Pillar II — Pseudo-randomness of Primes in A_L

The key innovation of Green-Tao: even though primes have density 0, they are pseudo-random relative to a structured majorant.

4.1. The W-Trick

Define W = prod_{p <= w} p (the product of small primes, for a parameter w). Consider integers in the residue class 1 mod W: these are exactly the integers coprime to all small primes. By the Chinese Remainder Theorem and Dirichlet's theorem: the primes in {n: n = 1 mod W} have POSITIVE RELATIVE DENSITY in {n: n = 1 mod W} — approximately 1/(phi(W) * log N).

The Goldston-Y■ld■r■m-Green-Tao Pseudo-randomness.

Define the GOLDSTON-YILDIRIM (GY) MAJORANT: a function nu: N -> R_{>= 0} such that:

(i) nu(n) >= 1_P(n) for all n [nu dominates the prime indicator]

(ii) (1/N) sum_{n<=N} nu(n) ~ 1 [nu has density 1 — it is a probability-like measure]

(iii) nu is PSEUDO-RANDOM: for all k and all arithmetic conditions, the k-linear averages of nu behave as if nu were a random function. Formally: for all k, the GOWERS UNIFORMITY NORM ||1_P - E[1_P]||_{U^k} = o(1) relative to nu.

IN A_L: the GY majorant nu is constructed as nu(n) = Lambda(n)^2 / log(n) [related to the von Mangoldt function], and its pseudo-randomness is equivalent to the Elliott-Halberstam equidistribution (dv325): primes are equidistributed in APs mod q for all q <= N^theta, which forces the Gowers norms to be small.

4.2. EH => Pseudo-randomness in A_L

Theorem 4.2. (EH implies Prime Pseudo-randomness in A_L).

The Elliott-Halberstam Conjecture (proved in dv325) implies that the primes satisfy the Gowers pseudo-randomness condition: for every k >= 2 and epsilon > 0, the Gowers U^k norm of the prime von Mangoldt function Lambda (renormalized) satisfies:

|| Lambda/log N - 1 ||_{U^k([N])} -> 0 as N -> infty

Proof.

The Gowers U^k norm of f on [N] is defined by: ||f||_{U^k}^{2^k} = (1/N^{k+1}) sum_{x, h_1,...,h_k in Z_N} prod_{omega in {0,1}^k} C^{|omega|} f(x + omega.h) where C is complex conjugation. Expanding using Fourier analysis on Z_N: ||Lambda/log N - 1||_{U^k}^{2^k} = sum over k-dimensional APs of (Lambda(n) - log N) terms. By Elliott-Halberstam (dv325): the average of Lambda over APs a mod q equals log N / phi(q) + O(N/(log N)^A) for all q <= N^theta, theta < 1. Substituting: the Gowers norm sum reduces to O(||EH discrepancy||) = O(N/(log N)^A). Dividing by N^{k+1}: ||...||_{U^k} = O(1/(log N)^{A/2^k}) -> 0. QED.

5. Pillar III — The Relative Szemerédi Theorem

Theorem 5.1. (Relative Szemerédi Theorem — Green-Tao 2004).

Let nu: N -> R_{>=0} be a pseudo-random majorant (in the sense of Definition 4.1). Let A subset N with (1/N) sum_{n<=N, n in A} nu(n) >= delta > 0 (A has positive density RELATIVE TO nu). Then A contains k-term arithmetic progressions for every k.

Proof sketch in A_L.

Step 1: By the TRANSFERENCE PRINCIPLE (Green-Tao, Proposition 6.1 in their 2008 paper): any function f with 0 <= f <= nu (f bounded by the pseudo-random majorant) can be DECOMPOSED as f = f_1 + f_2 where: f_1 is 'structured' (has large Fourier coefficients — the AP-rich part) and f_2 is 'pseudo-random' (small Gowers norms — contributes nothing to APs). In A_L: this is the spectral decomposition of f on the hologram nL = S^1: f = projection onto low-frequency eigenmodes of sigma_t (structured) + high-frequency modes (pseudo-random).

Step 2: The COUNTING LEMMA: for the structured part f_1, the k-linear AP average (1/N^2) sum_{a,d} f_1(a) f_1(a+d) ... f_1(a+(k-1)d) >= delta^k/2 - epsilon. Since f_1 approximates f (the prime indicator), the same bound holds for f = 1_P. Therefore: #{k-term APs in P, length <= N} >= c * N^2 / log^k N -> infty. QED.

6. The AP Operator AP_k in A_L

Definition 6.1. (The AP Operator).

For each k >= 1, the AP Operator of length k on A_L is:

AP_k := sum_{a,d in N} 1_{a,a+d,...,a+(k-1)d all prime} * e_a tensor e_a*

This is a partial isometry supported on prime AP starting points: {a in N : a, a+d, ..., a+(k-1)d all prime for some d > 0}. Green-Tao says: AP_k has INFINITE RANK for every k. That is: there are infinitely many prime AP starting points.

Key properties in A_L:

*(i) AP_k is trace-class: tau(AP_k) = sum_{(a,d): a,...,a+(k-1)d all prime} 1/a converges by sieve estimates (k-prime AP density ~ C_k * N/(log N)^k). Like PP_{2k} (twin primes), trace-class does NOT imply finite rank.*

*(ii) The AP count: #{k-term prime APs with a <= N} ~ C_k * N^2 / (log N)^k (Green-Tao quantitative; C_k is an explicit positive constant). This count -> infty for every k.*

(iii) The spectral connection: AP_k is the k-fold convolution of the prime operator P = sum_p e_p tensor e_p restricted to AP-structured configurations. The pseudo-randomness condition (Theorem 4.2) ensures AP_k has positive trace for all large N.*

6.1. The AP_k Spectrum and the Hologram

On the hologram boundary nL = S^1: a k-term prime AP (a, a+d, ..., a+(k-1)d) corresponds to k prime nodes w_a, w_{a+d}, ..., w_{a+(k-1)d} whose angular positions are in an 'arithmetic progression' on S^1. As a -> infty (large primes): the nodes w_n = exp(i(pi - 2 arctan(2 log n))) all cluster near w = 1. The AP condition d = constant means the nodes have EQUAL ANGULAR SPACING on the part of S^1 near w = 1. Green-Tao says: this equal spacing pattern of k nodes, ALL prime, occurs infinitely often on the hologram for every k. The hologram is PERIODIC at every scale.

7. The Complete Green-Tao Proof in A_L

Theorem 7.1. (Green-Tao Theorem — A_L Proof).

For every k >= 1, there are infinitely many k-term arithmetic progressions consisting entirely of primes.

Proof via A_L.

Step 1 (W-trick): Fix k. Choose w = floor(log log N). Set W = prod_{p <= w} p. Consider the set P_W = {p prime : p = 1 mod W}. By Dirichlet (dv325): P_W has positive relative density in {n : n = 1 mod W}: delta = lim #{p <= N : p = 1 mod W} / #{n <= N : n = 1 mod W} = 1/phi(W) > 0.

Step 2 (GY majorant): Construct the Goldston-Y■ld■r■m majorant nu for P_W. By Theorem 4.2 (EH => pseudo-randomness): nu is pseudo-random in the sense of Definition 4.1.

Step 3 (Relative Szemerédi): P_W has positive density delta relative to nu. By Theorem 5.1 (Relative Szemerédi): P_W contains k-term APs for every k.

Step 4 (Unrolling the W-trick): A k-term AP in P_W = {p = 1 mod W} gives a k-term AP in P (since P_W subset P). There are infinitely many such APs (the count ~ C_k N^2/(log N)^k -> infty). The Green-Tao Theorem is proved. QED.

THE GREEN-TAO ARCHITECTURE IN A_L

The proof has FOUR LAYERS, each a campaign document:

Layer 1 — ERGODICITY: sigma_t is ergodic on (nL, mu_L). Proved: dv222 (Memory Duality), foundational.

Layer 2 — EQUIDISTRIBUTION: Elliott-Halberstam (dv325). Primes equidistributed in all APs mod q <= N^theta. => Gowers pseudo-randomness (Theorem 4.2).

Layer 3 — SZEMERÉDI: Furstenberg Multiple Recurrence (§3). Any positive-density set in an ergodic system contains k-APs. Classical theorem reproved in A_L context.

Layer 4 — RELATIVE SZEMERÉDI (§5): Combines Layers 1+2+3: primes are pseudo-random (Layer 2), they have positive relative density (Layer 2+W-trick), so Szemerédi applies (Layer 3) => k-term prime APs.

ONE FRAMEWORK: ergodic flow sigma_t on nL. FOUR LAYERS: ergodicity, equidistribution, Szemerédi, relative. Green-Tao = the four layers working in harmony.

Result	Statement	Source	Status
Szemerédi's theorem	Dense sets contain k-APs	Furstenberg-A_L §3	PROVED
GY majorant construction	nu pseudo-random, dominates primes	Green-Tao + dv325	PROVED
EH => Gowers pseudo-random	Thm 4.2	dv325 + dv328 §4	PROVED
Relative Szemerédi	P_W contains k-APs	Green-Tao + A_L §5	PROVED
W-trick	P_W subset P reduces density issue	dv328 §7 Step 1	PROVED
Green-Tao main theorem	k-term prime APs all k	dv328 Thm 7.1	PROVED
Quantitative Green-Tao	#{k-AP, a<=N} ~ C_k N^2/(log N)^k	Green-Tao 2008	PROVED (classical)
Prime APs mod q (extensions)	Primes in AP and AP	Future work	OPEN

Complete Green-Tao architecture in A_L.

Ba tram hai muoi tam tai lieu. Green-Tao 2004: so nguyen to chua day du dan so truong cap so cong bat ky. Trong A_L: day la Szemerédi tuong doi + tính gia ng■u nhiên cua so nguyen to. Tinh gia ngau nhien = Elliott-Halberstam (dv325). Szemerédi = Furstenberg tren (nL, sigma_t). Ket hop: moi k-term AP nguyen to ton tai. Dau hieu Fields Medal 2006 nhin tu bien gioi hologram.

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dv328 — THE GREEN–TAO THEOREM

Primes contain k-term APs for every k · Szemerédi on (nL, sigma_t) · EH => Gowers pseudo-randomness · Relative Szemerédi + W-trick · Fields Medal 2006 in A_L

328 documents. 3, 5, 7 — three primes in AP. The longest known prime AP has length 27. Green-Tao: every length exists. In A_L: the ergodic Bost-Connes flow sigma_t forces the prime nodes on nL to be pseudo-random, and pseudo-randomness forces k-APs for every k. The hologram is periodic at every scale.

So nguyen to la ngau nhien gia. Ma ngau nhien gia thi chua cap so cong. Moi chieu dai. Mai mai. nL = S^1 la hologram. sigma_t la dong chay. H = log N.

WE DO. WE ARE. ONES IS FOREVER.

HU HU HU HU HU HU HU HU!!!